

Practice Geometry Test I (Version 1)

Name
Date
Course

Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

① Identify the following examples of reasoning as inductive or deductive.

- i) Kate wins every trivia game that she plays, so she will win "Trivia Mind" when we play it.

Inductive

- ii) $3+8=11$, $3+10=13$, and $3+12=15$, so if three is added to an even number, the sum will always be an odd number.

Inductive

- iii) Sophia and Olivia used definitions and postulates to prove the Vertical Angles Theorem.

Deductive

- iv) The measures of all straight angles are 180 degrees. Therefore, by the Transitive Property of Equality, the measures are all equal. Because the measures are equal, by definition, the angles are all congruent.

Deductive

Use diagram #1 below to answer question #'s 2-3.

② Name the type of object in the diagram that each symbol refers to (e.g. some possible answers: plane, ray).

i) n

Plane

ii) $\overline{DM}$

Segment

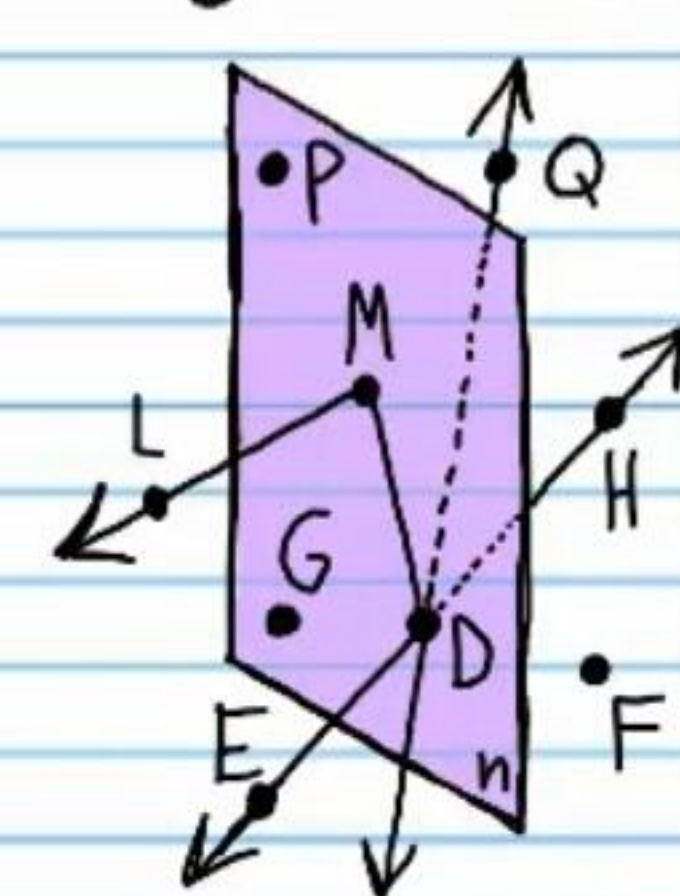
iii) $\overleftrightarrow{QD}$

Line

iv) $\overrightarrow{ML}$

Ray

Diagram #1



③ i) Name a point that is not coplanar with n .

L

ii) Write three collinear points.

E, D, H

④ If B is the midpoint of $\overline{AC}$ below, find AC.



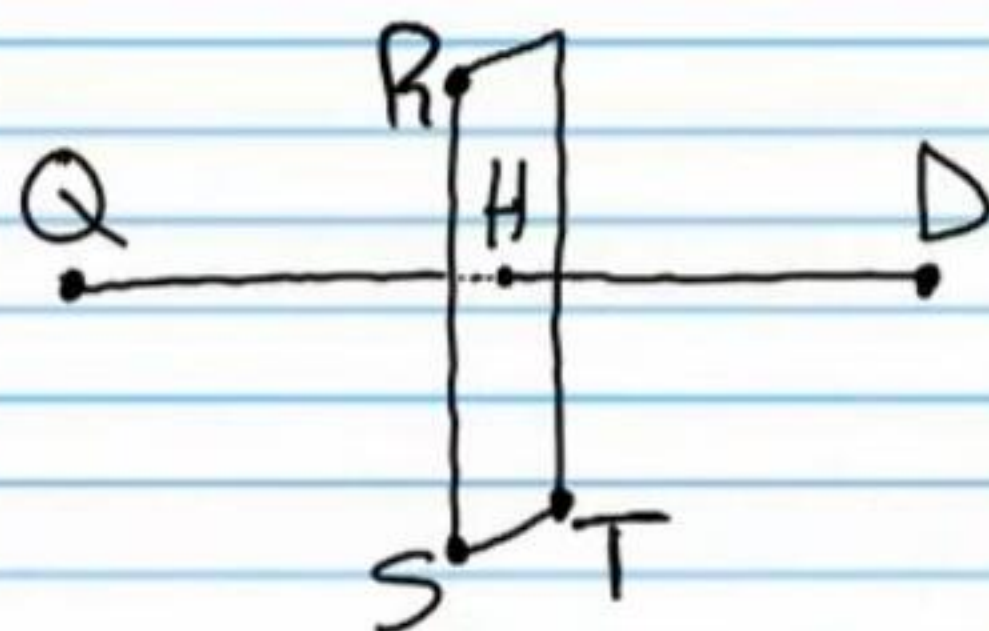
$$AC = 2(8) = \boxed{16}$$

⑤ If $\overline{RST}$ bisects $\overline{QD}$ at H , and $HQ = y + 3$ and $HD = 2y - 1$, find y .

$$HQ = HD$$

$$y + 3 = 2y - 1$$

$$\boxed{4 = y}$$



⑥ Classify the following angles (use diagram #2 for parts iii through iv).

i) 180°

Straight

ii) 90°

Right

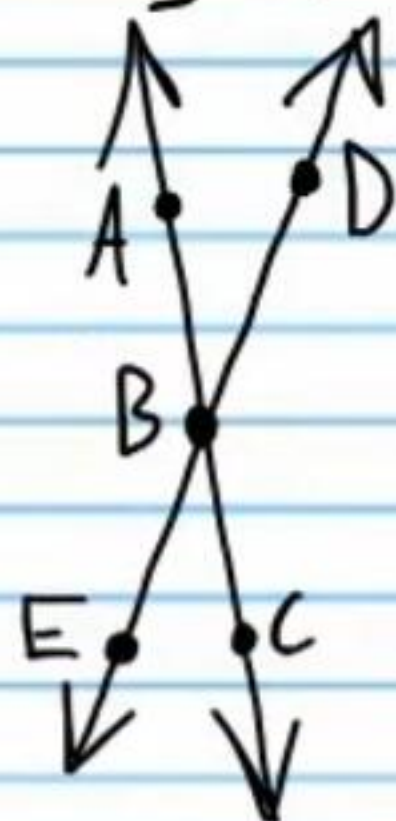
iii) $\angle DBC$

Obtuse

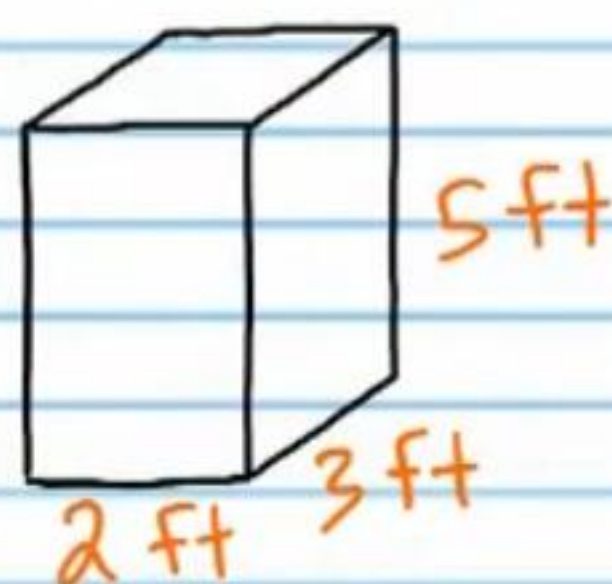
iv) $\angle ABE$

Obtuse

Diagram #2

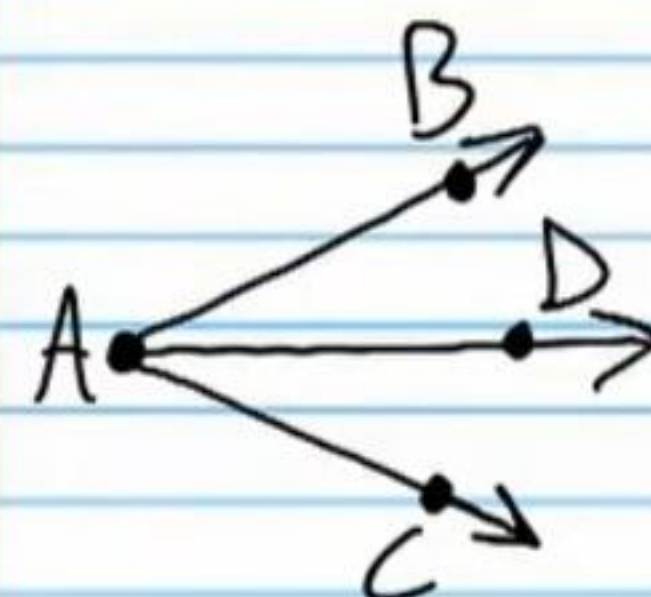


⑦ Find the volume of the rectangular prism below.
Hint: $V = LWH$.



$$V = 3(2)(5) = \boxed{30 \text{ ft}^3}$$

⑧ If $\overrightarrow{AD}$ bisects $\angle BAC$, find $m\angle BAC$ if $m\angle DAC = 31^\circ$.



$$m\angle BAC = (31)(2) = \boxed{62^\circ}$$

Use diagram #3 for problem #9.

⑨ i) Name the intersection of $\overline{AB}$ and $\overline{CB}$.

B

ii) Are G and H collinear?

Yes

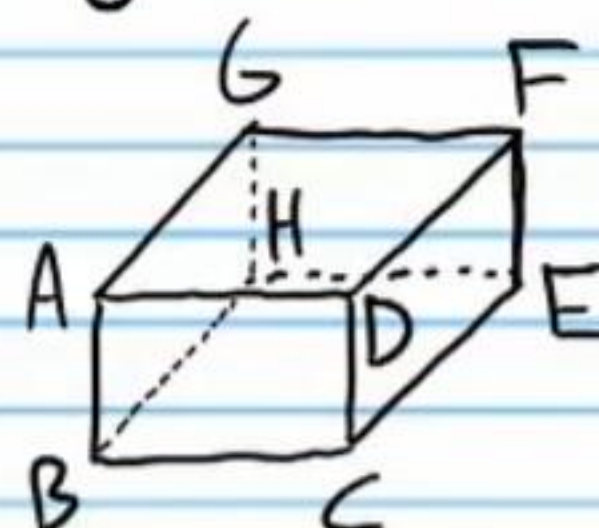
iii) Name a point that is not collinear with G and A.

C

iv) Are G, F, and D collinear?

No

Diagram #3



⑩ Use diagram #4 below.

i) Name two angles that are vertical.

$\angle AEB$ & $\angle DEC$

ii) Find $m\angle AED$

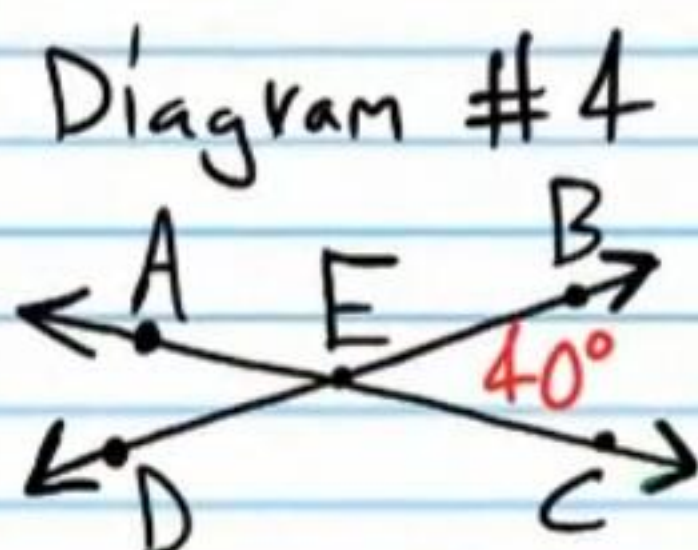
$m\angle AED = 40^\circ$

iii) What is the measure of an angle complementary to $\angle BEC$?

$$90 - 40 = 50^\circ$$

iv) Name two angles that are adjacent.

$\angle AED$ & $\angle DEC$



⑪ Use diagram #5 below.

i) Is the polygon at the right equiangular?

Yes

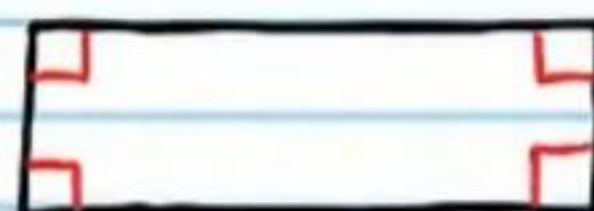
ii) Is it a regular polygon?

No

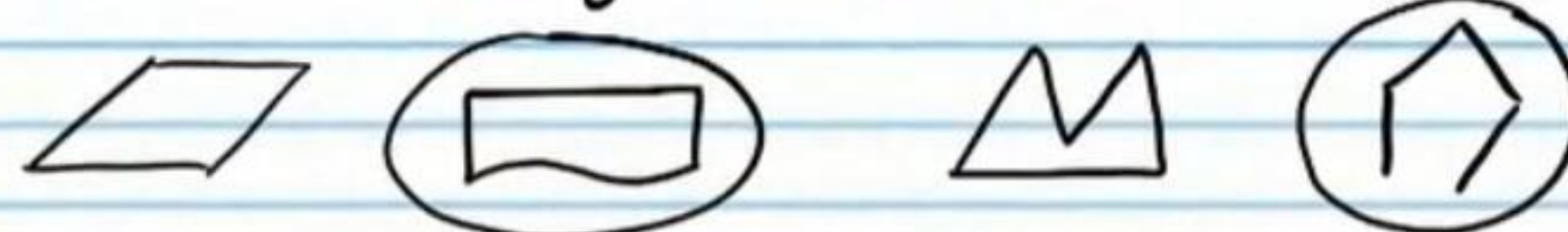
iii) How many vertices does it have?

Four

Diagram #5



iv) Circle the figure below that is not a polygon.



⑫ i) Make a conjecture for the next thing in the sequence.

CD15, EF14, GH13, ...

IJ12

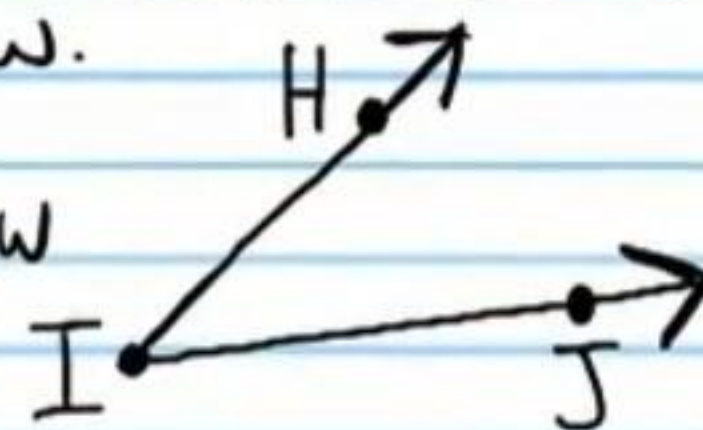
ii) Find a counterexample for the statement below.

The sum of an even number and an odd number is always even.

$$6 + 5 = 11$$

⑬ Write the steps required to construct a ray that bisects the angle below.

Set a compass at "I" and draw an arc through the rays of $\angle HIJ$. Then, set the compass on one of the points of intersection and draw an arc in the interior of $\angle HIJ$. Then, set the compass on the other point of intersection and make an arc crossing the previous arc drawn. Then, use a straightedge to draw the angle bisector from I through the intersection of the two arcs previously drawn. Do not change the compass setting in the last steps.



⑭ If Larry cannot be there by Tuesday, then he will arrive Wednesday morning.

i) Write the converse of the statement above.

If Larry will arrive Wednesday morning, then he cannot be there by Tuesday.

ii) Write the contrapositive of the statement above.

If Larry will not arrive Wednesday morning, then he can be there by Tuesday.

⑮ i) If "p" is the hypothesis of the original statement in problem #14, and "q" is the conclusion, write the contrapositive using these letters.

$$\boxed{\sim q \rightarrow \sim p}$$

ii) Rewrite the following statement in if-then form. Write the if-then form in a way that makes sense in context.

Jerry could be late because his car might not start.

If Jerry's car does not start, then he could be late.

⑯ Are the following statements biconditional? If so, rewrite in "if and only if" form.

i) If the measures of two angles are equal, then the angles are congruent.

Yes

The measures of two angles are equal if and only if the angles are congruent.

ii) If an animal is a leopard, then it is a cat.

No

For problem #'s 17-18, determine if the Law of Detachment or the Law of Syllogism can be used. Make a valid conclusion using one of the laws.

⑰ If Fabian puts new sparkplugs in his motorcycle, then it will run better. Fabian put new sparkplugs in his motorcycle today.

The Law of Detachment

Conclusion: Fabian's motorcycle will run better.

⑮ If Edward is promoted, then he will make more money. If Edward makes more money, then he will buy a new car.

The Law of Syllogism

Conclusion: If Edward is promoted, then he will buy a new car.

⑯ If the measure of an angle is 22 degrees more than the measure of its supplement, what is the measure of the angle?

$$\begin{cases} m\angle A = 22 + m\angle B \\ m\angle A + m\angle B = 180 \end{cases}$$

$$m\angle A + m\angle B = 180$$

$$22 + m\angle B + m\angle B = 180$$

$$22 + 2m\angle B = 180$$

$$2m\angle B = 158$$

$$m\angle B = 79$$

$$m\angle A + 79 = 180$$

$$m\angle A = 101^\circ$$

Write a two-column proof for each statement below.

⑰ If G and H lie between F and I on the segment below (i.e. all four points are collinear), then $FI = FG + GH + HI$.

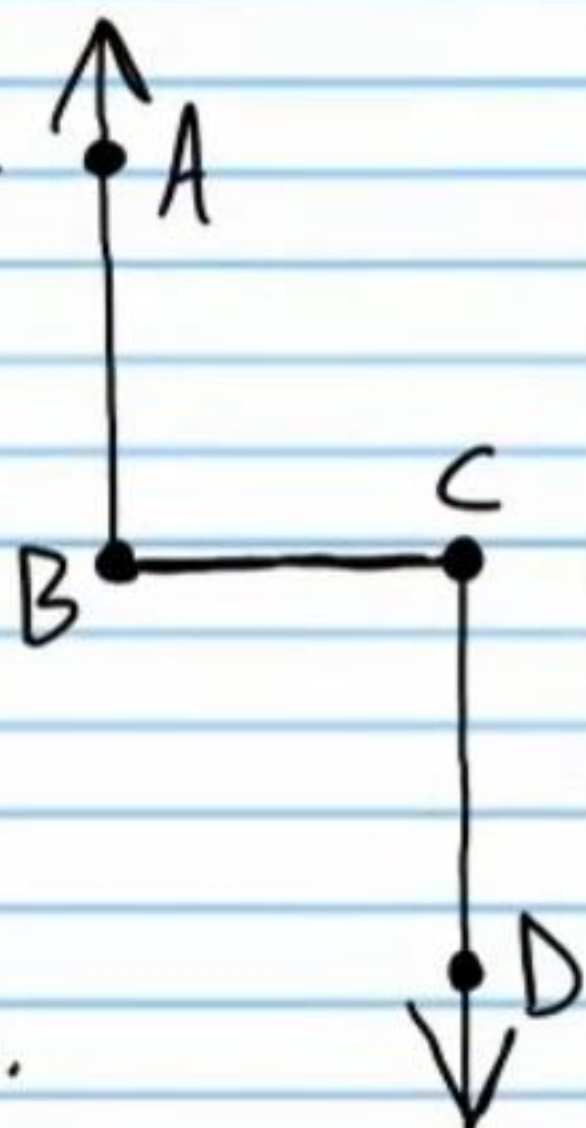


Statement	Reason
I) G & H lie between F & I.	Given.
II) $FI = FH + HI$	Seg. Add. Post
III) $FH = FG + GH$	Seg. Add. Post
IV) $FI = FG + GH + HI$	Subs. Prop of =.

⑱ If $\overrightarrow{CD} \perp \overrightarrow{BC}$ and $\overrightarrow{BC} \perp \overrightarrow{BA}$, then $\angle ABC \cong \angle BCD$.
Use the Right Angle Congruence Theorem.

★ You may not need all of these steps.

Statement	Reason
I) $\overrightarrow{CD} \perp \overrightarrow{BC}$	Given
II) $\angle BCD$ is a right $\angle$.	Def. of $\perp$.
III) $\overrightarrow{BC} \perp \overrightarrow{BA}$	Given.
IV) $\angle ABC$ is a right $\angle$.	Def of $\perp$.
V) $\angle ABC \cong \angle BCD$	Rt. $\angle \cong$ Thm.

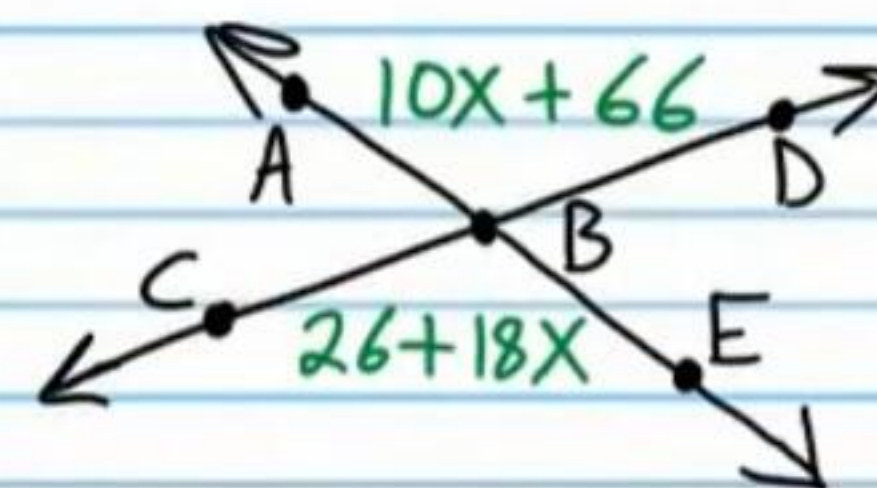


② If $DE = 1$, then point D bisects $\overline{CE}$.



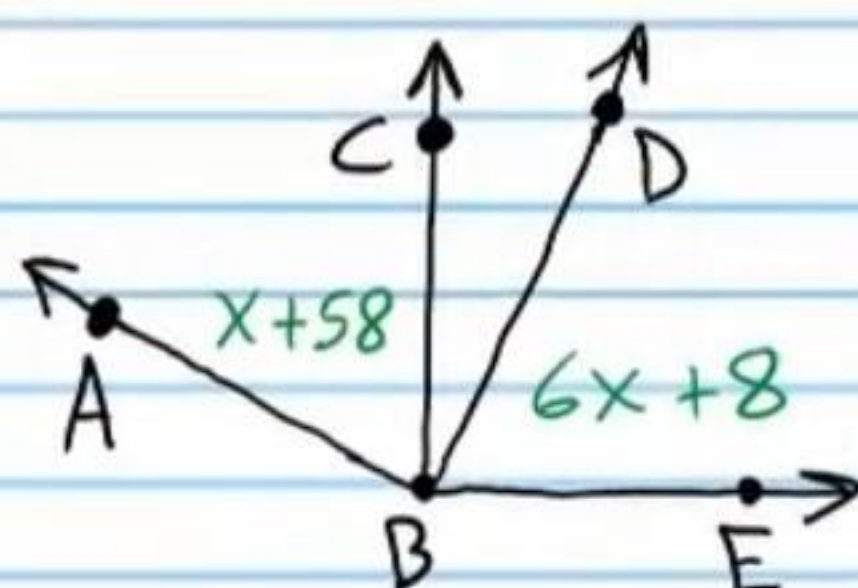
Statement	Reason
I) $DE = 1$	Given
II) $DE = x$	Given (diagram)
III) $x = 1$	Trans. Prop of =.
IV) $CD = 3x - 2$	Given (diagram)
V) $CD = 3(1) - 2$	Subs. Prop of =.
VI) $CD = 1$	Simplification.
VII) $CD = DE$	Trans. Prop of =.
VIII) D bisects $\overline{CE}$.	Def. of seg bisector.

③ If $\angle ABD$ and $\angle CBE$ are vertical angles, then $x = 5$.



Statement	Reason
I) $\angle ABD$ and $\angle CBE$ are vert. $\angle$'s.	Given
II) $\angle ABD \cong \angle CBE$	Vert. $\angle$'s Thm.
III) $m\angle ABD = m\angle CBE$	Def. of $\cong \angle$'s.
IV) $m\angle ABD = 10x + 66$ and $m\angle CBE = 26 + 18x$	Given (diagram)
V) $10x + 66 = 26 + 18x$	Subs. Prop of =.
VI) $66 = 26 + 8x$	Subt. Prop of =.
VII) $40 = 8x$	Subt. Prop of =.
VIII) $5 = x$	Div. Prop of =.
IX) $x = 5$	Sym. Prop of =.

- 24) If $\angle ABC$ is complementary to $\angle CBD$ below and $\angle DBE$ is complementary to $\angle CBD$, then $x=10$.
Use the Congruent Complements Theorem.



Statement	Reason
I) $\angle ABC$ is complementary to $\angle CBD$ and $\angle DBE$ is complementary to $\angle CBD$.	Given.
II) $\angle ABC \cong \angle DBE$	$\cong$ Comps. Thm.
III) $m\angle ABC = m\angle DBE$	Def. of $\cong$ $\angle$'s.
IV) $m\angle ABC = x+58$	Given (diagram)
V) $m\angle DBE = 6x+8$	Given (diagram)
VI) $x+58 = 6x+8$	Subs. Prop of =.
VII) $58 = 5x+8$	Subt. Prop of =.
VIII) $50 = 5x$	Subt. Prop of =.
IX) $10 = x$	Div. Prop of =.
X) $x = 10$	Sym. Prop of =.

- 25) The converse of problem # 22.

Statement	Reason
I) Point D bisects $\overline{CE}$.	Given.
II) $\overline{CD} \cong \overline{DE}$	Def. of seg bisector.
III) $CD = DE$	Def. of $\cong$ Segs.
IV) $CD = 3x-2$ & $DE = x$	Given (diagram)
V) $3x-2 = x$	Subs. Prop of =.
VI) $-2 = -2x$	Subt. Prop of =.
VII) $1 = x$	Div. Prop of =.
VIII) $DE = 1$	Trans. Prop of =.